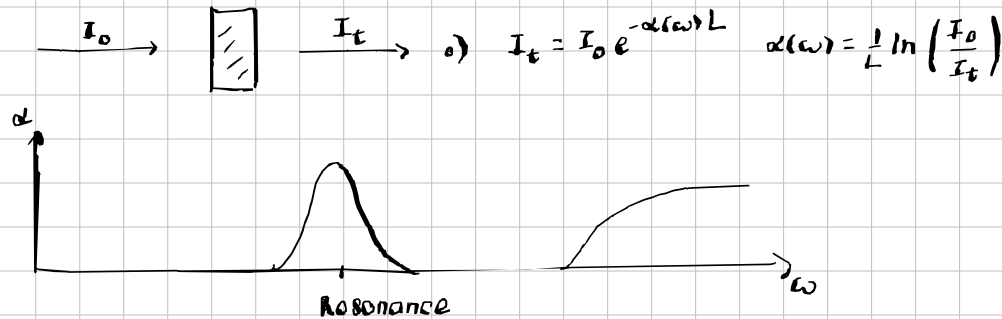


Why should we do solid states?

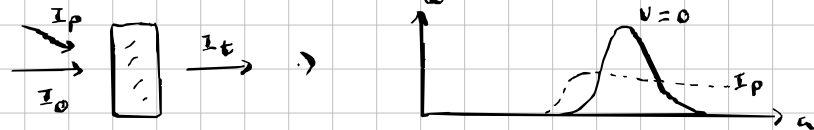
In nature, we usually observe linear absorption spectrum



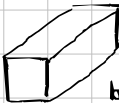
We can modify the absorption spectrum with a voltage. The electric field changes exciton's peak



Some modify with ultrafast laser

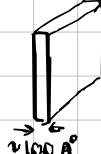


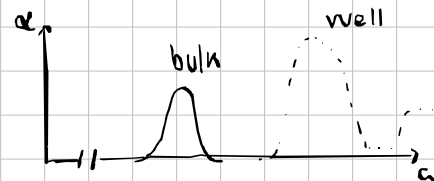
Some modify with quantum confinement

 bulk: a very small nano scale can be considered bulk.
bulk has many electrons $\rightarrow$ band
Electron's wavelength is not confined

 Quantum wire
 Quantum dot

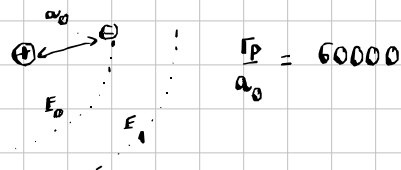
Wire confines in two dimensions
Dot confines in three dimensions

 Quantum well
Electron is confined vertically

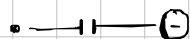


By Confinement, what is confined? The answer is interesting.

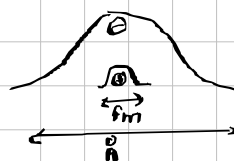
Consider a Hydrogen atom, Bohr radius is $a_0 = 0.529 \text{ nm} = 0.529 \text{ \AA}$. Meanwhile, proton has radius $r_p \approx 0.824 \text{ fm}$



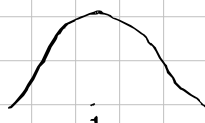
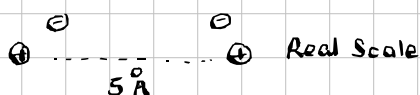
Atom is mostly empty



Wave function $\psi(x)$



In solid, the distance between atom is $\approx 0.5 \text{ nm}$. Even when people say electron is "small", yes it is small in mass, but its probability spreads over $\approx \frac{1}{100} \text{ nm}$

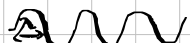


Electron does not have a single momentum, it has a bandwidth

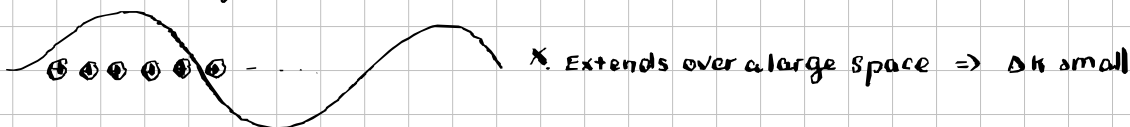
$$\lambda = \frac{h}{p} \Rightarrow \text{There is a band width in wavelength}$$

$$p = mv \quad v = 10^6 \text{ m/s} \quad \lambda = 0.73 \text{ nm} \Rightarrow \text{This is roughly over 10 atoms in a lattice}$$

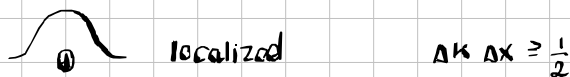
$$= \hbar k \quad \vec{v} = \frac{\hbar}{m} \vec{k}$$



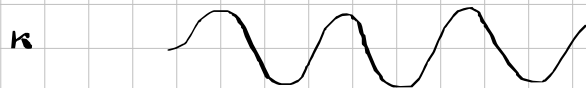
• An electron can have nearly one momentum: small bandwidth. This is solid state



• An electron can have a large momentum bandwidth



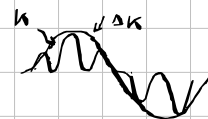
The bandwidth creates an envelope. This is why, we see



$$\Delta k, k \quad \psi_1 + \psi_2 + \dots = \text{localized}$$

$$\Delta k \uparrow \rightarrow \text{localized}$$

Somewhere in middle



When we say confinement, what happens is actually allowing many k or λ where

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad \lambda_n = \frac{2L}{n} \quad \psi_1 + \psi_2 + \dots =$$

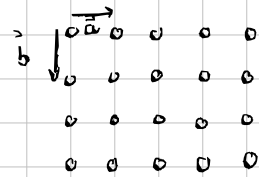
The superposition is confined. The electron is in a superposition of different possible momenta!

Note

$$k, \Delta k \text{ electron} \neq \text{a superposition electron of different } k$$

Cubic Basics

Consider the lattice, solid basically mean atoms or ions in periodic arrangement.



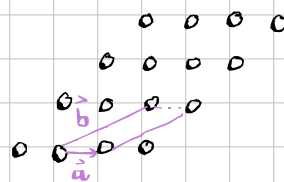
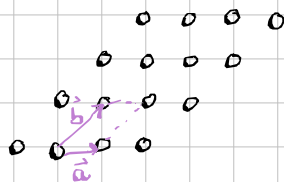
Ideal lattice positions follow Bravais

$$\vec{R}_n = n_1 \vec{a} + n_2 \vec{b} + n_3 \vec{c}$$

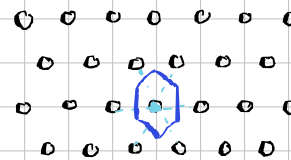
where n are integers and $\vec{a}$, $\vec{b}$, and $\vec{c}$ are primitive vectors.

Primitive means $\Omega_c = \vec{a} \cdot (\vec{b} \times \vec{c}) = \text{volume is minimized}$

Primitives are not unique



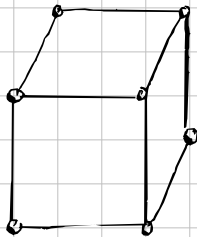
Wigner-Seitz Primitive cell



Primitive vectors create the lattice not the cell

The primitive cell needs:

tile all space with translation
no gaps/overlaps. Contains one exactly lattice point

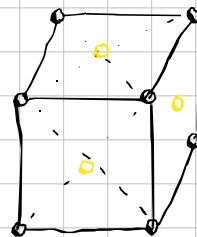


Simple Cubic

Each corner is shared with 8 cubes

$$N = 8 \cdot \frac{1}{8} = 1$$

The cell is primitive a^3



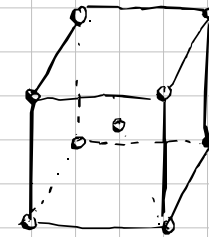
Face centered Cubic

FCC

$$N = 8\left(\frac{1}{8}\right) + 6\left(\frac{1}{2}\right)$$

$$= 4$$

The primitive cell is $a^3/4$



Body centered cubic

BCC

$$N = 8\left(\frac{1}{8}\right) + 1 = 2$$

The cell is primitive as $a^3/2$

Basis each lattice point has a basis which might be multiple atoms per point or complex molecules

$$\vec{r}_0 = \vec{r}_0$$

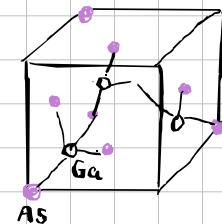
$$\vec{r}_0 = \vec{r}_0$$

Crystal Structure = Lattice + Basis $\vec{r}_{n,\alpha} = \vec{R}_n + \vec{r}_\alpha \quad \alpha = 1, 2, 3$

GaAs: zincblende $\vec{r} = \frac{1}{4}(a, a, a)$

$$\text{FCC Ga} = (0, 0, 0) \quad \text{As} = \frac{1}{4}(1, 1, 1)a$$

This two fcc means GaAs can be pictured as two identical FCC slightly shifted



Reciprocal Lattice

Let $\rho(\vec{r})$ be electronic charge density at position $\vec{r}$, a lattice translation

$$\rho(\vec{r} + \vec{R}_n) = \rho(\vec{r}) \approx |\psi(\vec{r})|^2$$

This density should be identical by symmetry.

Write

$\rho(\vec{r}) = \sum_{\vec{G}} S(\vec{G}) e^{i\vec{G} \cdot \vec{r}}$ which is simply sum of plane waves
 $\vec{G}$ is spatial frequency $\lambda = \frac{2\pi}{|\vec{G}|}$ $S(\vec{G}) = \frac{1}{\Omega_c} \int \rho(\vec{r}) e^{-i\vec{G} \cdot \vec{r}} d^3r$

A lattice translation says

$$\rho(\vec{r} + \vec{R}) = \sum_{\vec{G}} S(\vec{G}) e^{i\vec{G} \cdot (\vec{r} + \vec{R})} \Rightarrow \vec{G} \cdot \vec{R} = 2\pi N$$

$$\vec{R} = n_1 \vec{a} + n_2 \vec{b} + n_3 \vec{c} \quad \vec{R} = n_1 \vec{a} + n_2 \vec{b} + n_3 \vec{c}$$

$$\vec{A} \cdot \vec{a} = \vec{B} \cdot \vec{b} = \vec{C} \cdot \vec{c} = 2\pi$$

This says only certain electron momentum or wave number shift are allowed to preserve the state.

To find $\vec{A}, \vec{B}, \vec{C}$,

$$\Omega = \vec{a} \cdot (\vec{b} \times \vec{c}) \quad \vec{A} = \frac{2\pi}{\Omega} (\vec{b} \times \vec{c}) \quad \vec{B} = \frac{2\pi}{\Omega} (\vec{c} \times \vec{a}) \quad \vec{C} = \frac{2\pi}{\Omega} (\vec{a} \times \vec{b})$$

If the primitive cell in reciprocal lattice looks like Wigner Seitz, it is Brillouin zone